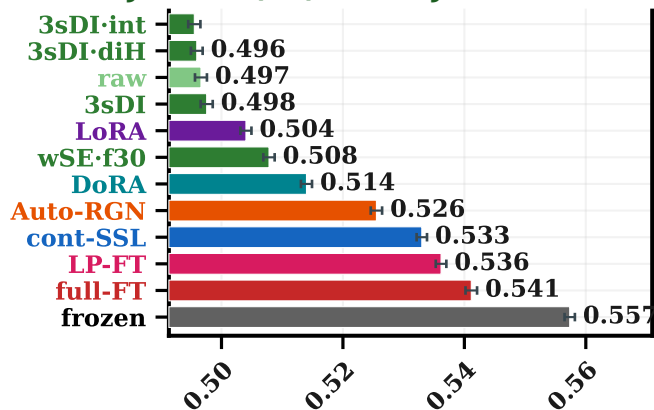


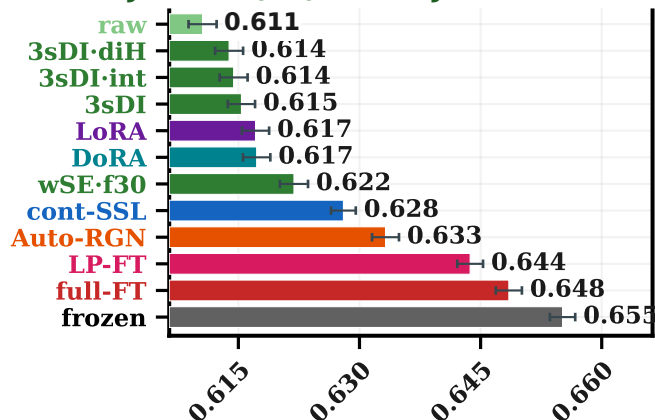
FactorJEPA vs. every fine-tuning rival: the four predictive metrics it wins

(Stratified 10k · n_test = 1,825 · 95% BCa CI)

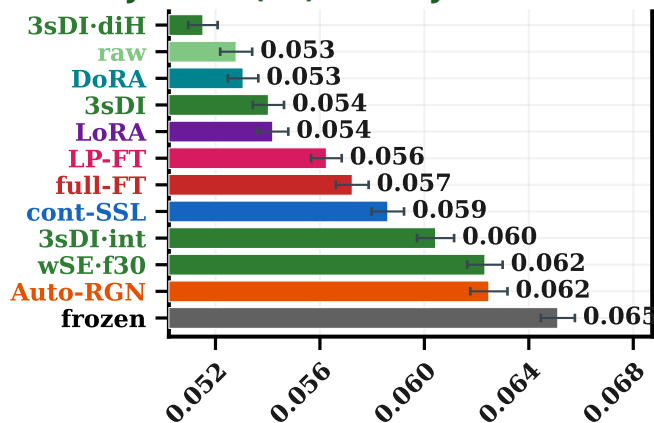
future-frame L1 ↓
vJEPA2.1 (2B): FactorJEPA wins +6.3×CI



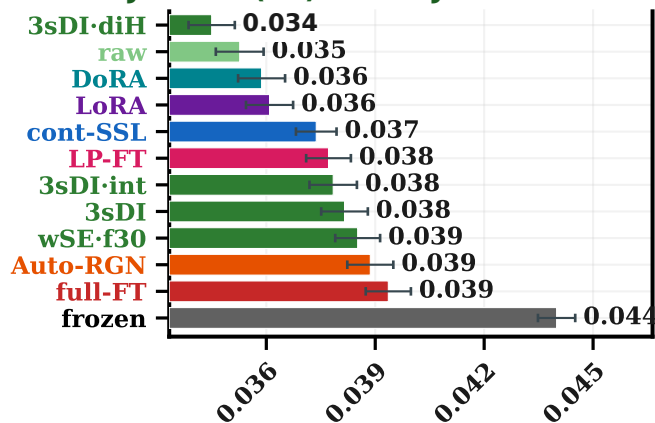
causal future-block L1 ↓
vJEPA2.1 (1B): FactorJEPA wins +2.7×CI



mask-ratio robustness slope ↓
vJEPA2.1 (1B): FactorJEPA wins +1.9×CI



free-running exposure-bias gap ↓
vJEPA2.1 (1B): FactorJEPA wins +1.5×CI



ENCODER KEY · short code = full name · ↓ = lower is better (all four are error metrics)
green shades = FactorJEPA / OURs · other colours = fine-tuning rivals
DI = interaction factor D_I · 3-stage = Layout → Agent → Interaction surgery

■ 3sDI = FactorJEPA · 3-stage (L→A→interaction)
■ 3sDI-diH = FactorJEPA · 3-stage (interaction-heavy)
■ 3sDI-int = FactorJEPA · 3-stage + intervention
■ raw = FactorJEPA-RAW (no factor targets)
■ wSE·f30 = FactorJEPA + WiSE-FT merge 30%
■ Auto-RGN = Auto-RGN gradient-norm surgery

■ DoRA = DoRA (weight-decomposed LoRA)
■ LP-FT = LP-FT (linear-probe then fine-tune)
■ LoRA = LoRA (low-rank adaptation)
■ cont-SSL = continual SSL (V-JEPA pretrain)
■ frozen = frozen V-JEPA 2.1 encoder
■ full-FT = full fine-tuning